

Real-Time Data Pipeline for Smart Irrigation Systems

Abstract

Water scarcity is one of the main global challenges in the 21st century, driven by increasing demands on freshwater resources for agriculture. An efficient irrigation system requires an accurate management to avoid using more water than necessary, increasing water productivity and reducing energy pumping costs. Optimization of irrigation systems does not rely solely on precise control, but also involves continuous real-time monitoring of crop conditions. Deployed sensors across the field measure soil, plant and environmental variables, generating vast amounts of data collected daily. The analysis of such data enables actionable insights towards enhanced agriculture performance. This work proposes a data pipeline architecture that can be applied to olive orchard irrigation, describing every step in the process of knowledge discovery. From the extraction and integration of sensor data, through processing, to visualization and analysis. It aims to support decision-making by inferring plant water requirements via the estimation of tree transpiration based on field data, while uncovering meteorological, soil and physiological hidden patterns with the aid of machine learning models.

Keywords: Data-driven Irrigation; Data Mining; Knowledge Discovery; Machine Learning.

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